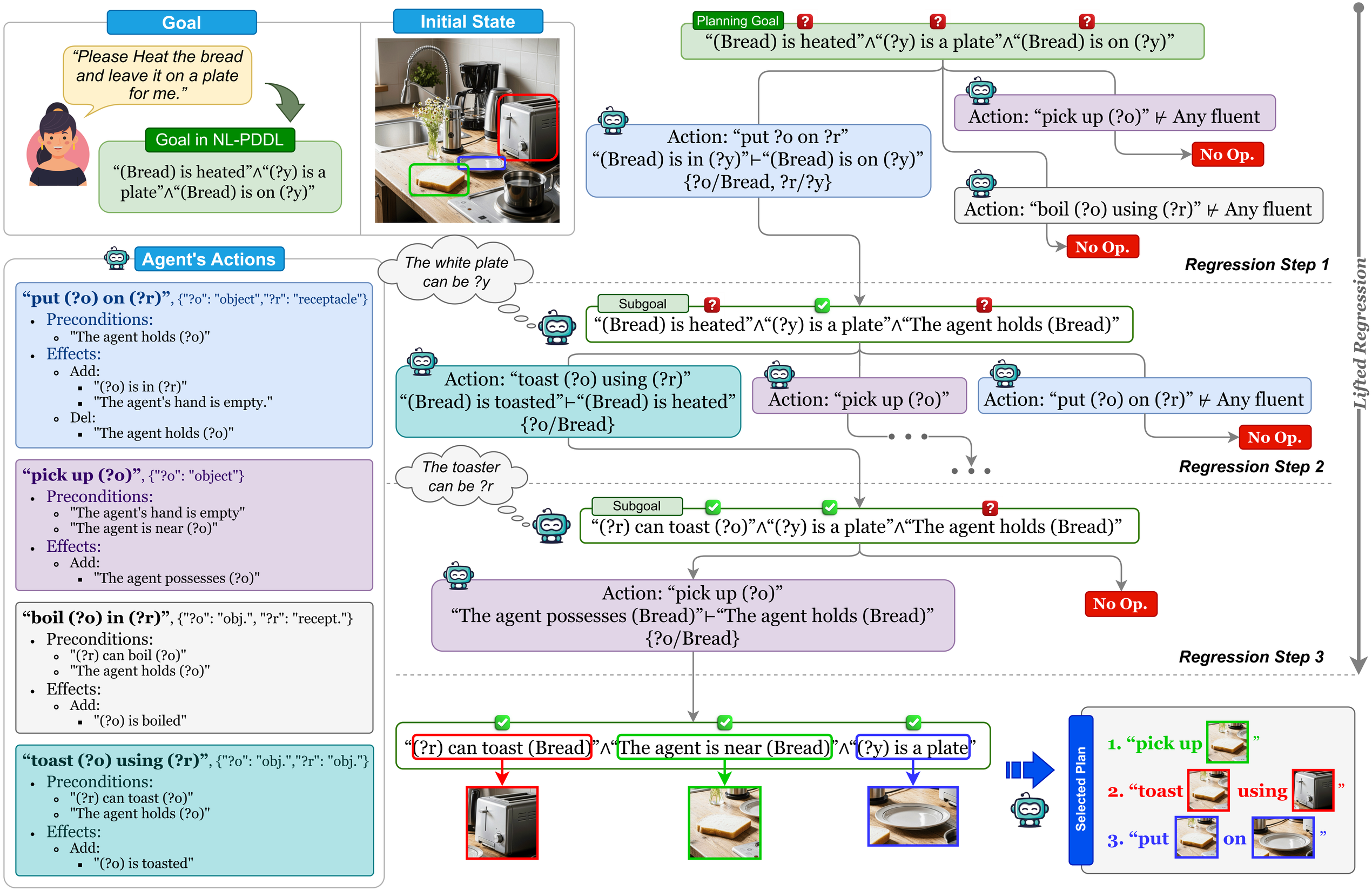
Goal
"Please Heat the bread and leave it on a plate for me."
Goal in NL-PDDL
"(Bread) is heated"∧"(?y) is a plate"∧"(Bread) is on (?y)"
Initial State
Agent's Actions
"put (?o) on (?r)", {"?o": "object","?r": "receptacle"}
Preconditions:
"The agent holds (?o)"
Effects:
Add:
"(?o) is in (?r)"
"The agent's hand is empty."
Del:
"The agent holds (?o)"
"pick up (?o)", {"?o": "object"}
Preconditions:
"The agent's hand is empty"
"The agent is near (?o)"
Effects:
Add:
"The agent possesses (?o)"
"boil (?o) in (?r)", {"?o": "obj.", "?r": "recept."}
Preconditions:
"(?r) can boil (?o)"
"The agent holds (?o)"
Effects:
Add:
"(?o) is boiled"
"toast (?o) using (?r)", {"?o": "obj.","?r": "obj."}
Preconditions:
"(?r) can toast (?o)"
"The agent holds (?o)"
Effects:
Add:
"(?o) is toasted"
Planning Goal
"(Bread) is heated"∧"(?y) is a plate"∧"(Bread) is on (?y)"
Action: "put ?o on ?r"
"(Bread) is in (?y)"⊢"(Bread) is on (?y)"
{?o/Bread, ?r/?y}
Action: "pick up (?o)" ⊬ Any fluent
No Op.
Action: "boil (?o) using (?r)" ⊬ Any fluent
No Op.
Regression Step 1
The white plate can be ?y
Subgoal
"(Bread) is heated"∧"(?y) is a plate"∧"The agent holds (Bread)"
Action: "toast (?o) using (?r)"
"(Bread) is toasted"⊢"(Bread) is heated"
{?o/Bread}
Action: "pick up (?o)"
Action: "put (?o) on (?r)" ⊬ Any fluent
No Op.
Regression Step 2
The toaster can be ?r
Subgoal
"(?r) can toast (?o)"∧"(?y) is a plate"∧"The agent holds (Bread)"
Action: "pick up (?o)"
"The agent possesses (Bread)"⊢"The agent holds (Bread)"
{?o/Bread}
No Op.
Regression Step 3
Lifted Regression
"(?r) can toast (Bread)"∧"The agent is near (Bread)"∧"(?y) is a plate"
Selected Plan
1. "pick up "
2. "toast using "
3. "put on "